Supplement: Solving $\psi(r) = 0$
A Short Guide to Root Finding

Kiara (Jihyun) Kim

For HW3 Question B4

Econ 4320

Where This Comes Up

In HW3 Question B4 you recover $\hat{F}_V, \hat{f}_V$ and then find the optimal reserve price r that solves

$$\psi(r) = 0, \qquad \psi(v) \equiv v - \frac{1 - F_V(v)}{f_V(v)}.$$

$\psi(r) = 0$ is an equation. Finding the r that satisfies it is a job called root finding.

This short deck explains what a root finder is, how it works, and how to call one.

Two Ways to Find the Root: Minimize vs. Solve

- Optimization: find the x that makes $f(x)$ as *small* as possible (close to 0).
  - To find a root this way you would minimize $f(x)^2$, whose smallest value, 0, sits exactly at the root.
  - Tools: `fmincon` (MATLAB), `scipy.optimize.minimize`, `optim` (R).
  - Walks "downhill" (against the slope) and stops at a flat spot, where the slope is 0.

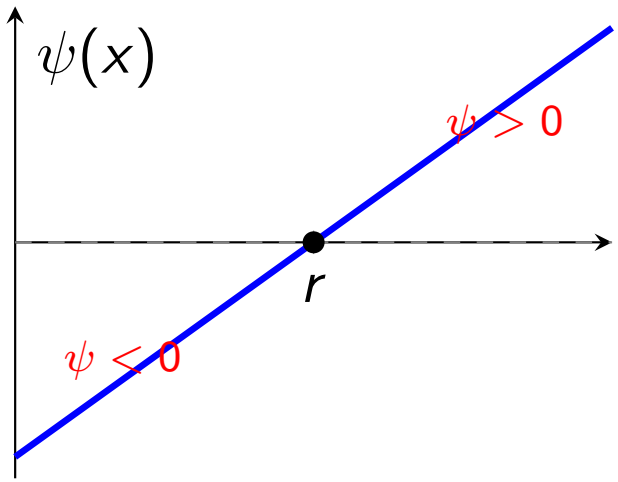
- Root finding: find the x where $f(x) = 0$ *directly* (a *crossing*).
  - Tools: `fzero` (MATLAB), `brentq`/`fsolve` (Python), `uniroot` (R).

Minimizing $\psi(r)^2$ also works, but it is indirect: slower, less precise, and it can settle in a dip where $\psi \neq 0$.

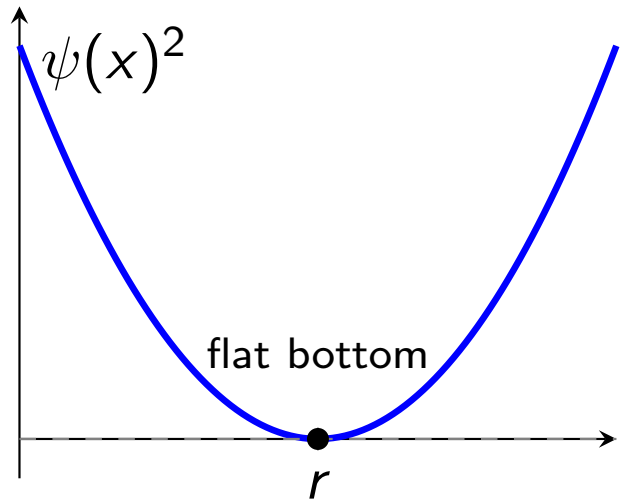
Why Not Just Minimize?

Left: a root finder uses the sign change of ψ. Right: minimizing ψ^2 only sees a flat valley that touches zero.

solve: ψ crosses 0



minimize: ψ^2 touches 0

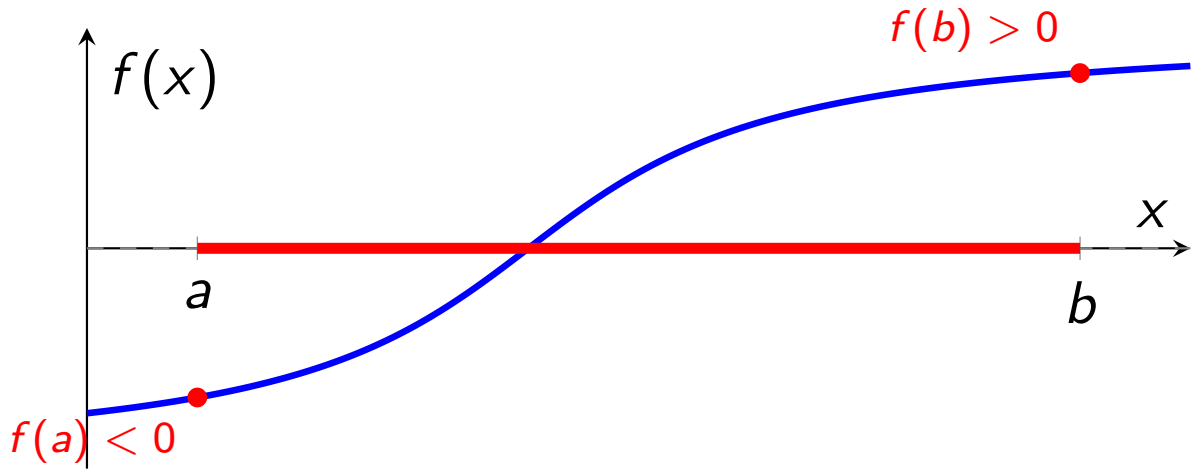


The sign flip (left) brackets the root and guarantees convergence. The squared version (right) is always ≥ 0 and merely *touches* zero, so a minimizer crawls over a flat bottom and can stop short of the true root.

The Idea: Bisection

Start with a bracket $[a, b]$ where $f(a)$ and $f(b)$ have opposite signs. Then a crossing must lie in between (Intermediate Value Theorem).

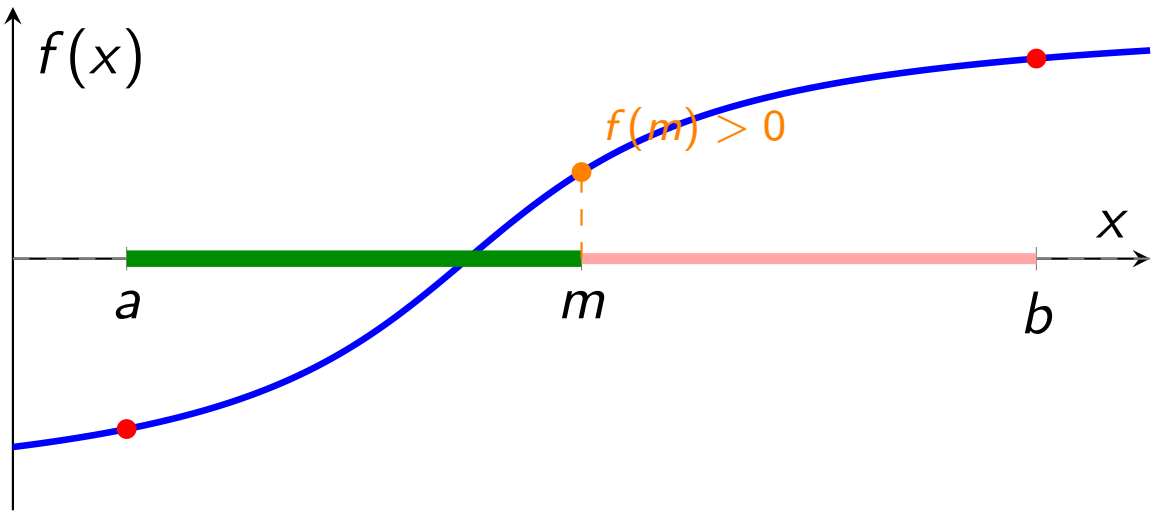
Evaluate the midpoint $m = \frac{a+b}{2}$; keep whichever *half* still shows a sign change. The bracket halves every step and closes in on the root. We trace it out on the next slides, starting from $[a, b] = [0.1, 0.9]$ (root at 0.4).



Opposite signs at the ends (red bracket): the zero is trapped inside.

Bisection, Step 1: [0.1, 0.9]

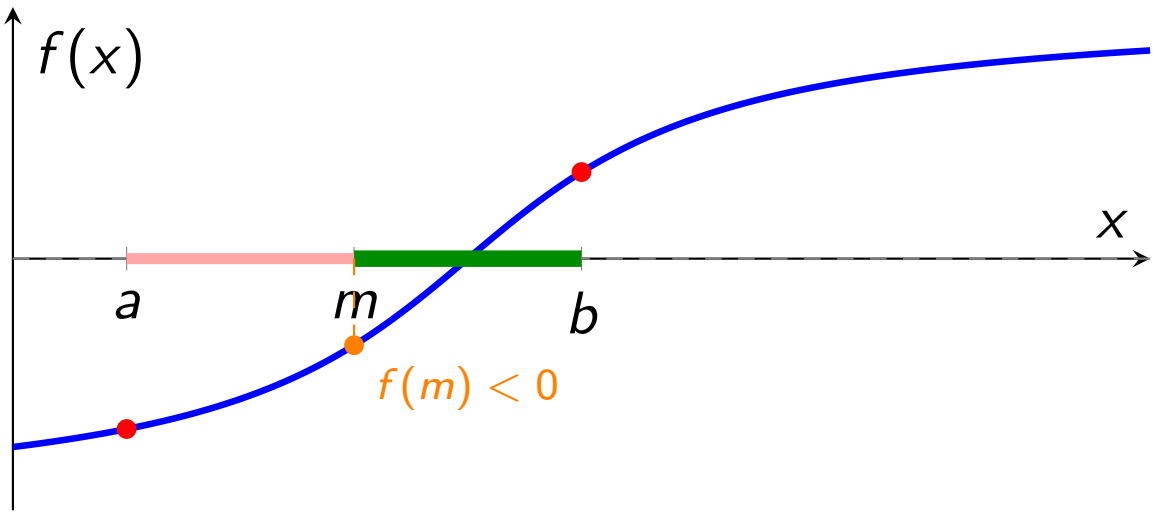
Midpoint $m = \frac{0.1+0.9}{2} = 0.5$.
Check its sign: $f(0.5) > 0$.



$f(m)$ has the *same* sign as $f(b)$, so the crossing is in the left half.
Keep $[a, m] = [0.1, 0.5]$ (green); discard $[m, b]$.

Bisection, Step 2: [0.1, 0.5]

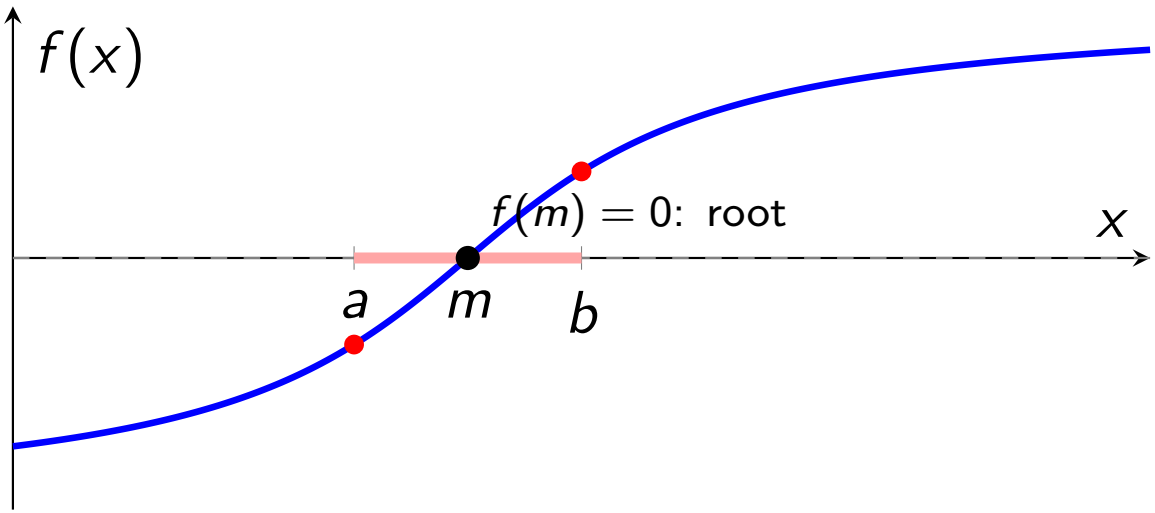
New midpoint $m = \frac{0.1+0.5}{2} = 0.3$.
Check its sign: $f(0.3) < 0$.



$f(m)$ now matches $f(a)$, so the crossing is in the right half.
Keep $[m, b] = [0.3, 0.5]$ (green).

Bisection, Step 3: [0.3, 0.5]

New midpoint $m = \frac{0.3+0.5}{2} = 0.4$.
Check its sign: $f(0.4) = 0$.



The midpoint *is* the root. In general you stop once the bracket is narrower than your tolerance and report m.

Why a Bracket, and Why Monotonicity Helps

- You must bracket the root: pick a, b with $f(a) < 0 < f(b)$. Opposite signs *guarantee* a zero inside.
- Monotone $f \Rightarrow$ exactly one crossing. Under the usual regularity condition ψ is increasing, so there is a single root and the finder cannot land on the wrong one.
- A bad bracket fails: if $f(a)$ and $f(b)$ have the *same* sign, the solver errors out. Choose a, b *inside the trimmed support*, where $\hat{f}_V$ in the denominator is reliable.

In Code

Python (scipy):

```
from scipy.optimize import brentq
def psi_hat(v):
    F = my_ecdf(v, vhat)          # your CDF of
        pseudo-values
    f = my_epdf(v, vhat, h)       # your kernel
        density
    return v - (1 - F) / f
r_hat = brentq(psi_hat, lo, hi) # psi_hat(lo) <
    0 < psi_hat(hi)
```

R:

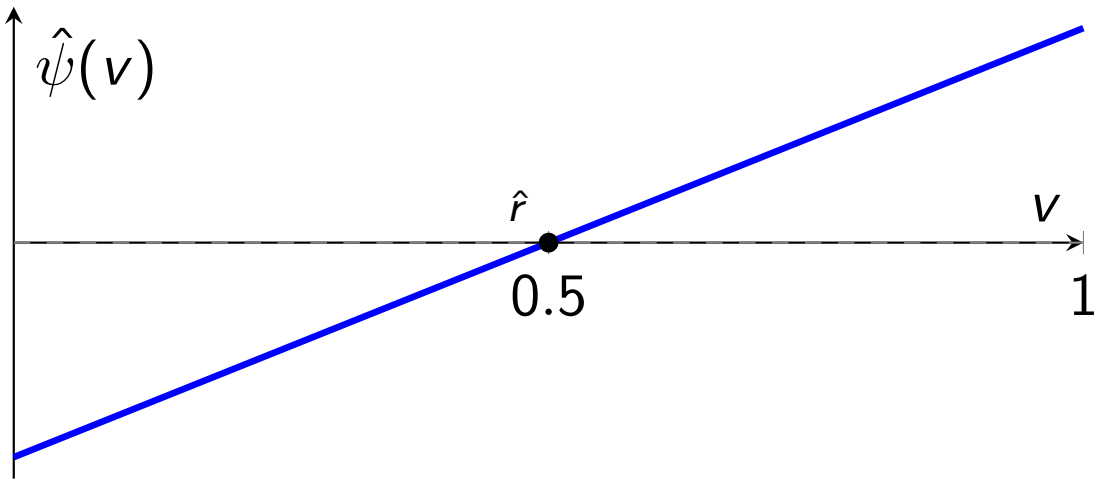
```
psi_hat <- function(v) v - (1 - my_ecdf(v, vhat)
    ) / my_epdf(v, vhat, h)
r_hat   <- uniroot(psi_hat, c(lo, hi))$root
```

MATLAB would be `fzero(@psi_hat, [lo hi])` — *not* `fmincon`.

Back to the Reserve Price

Build $\hat{\psi}$ from your estimates and look for the crossing:

$$\hat{\psi}(v) = v - \frac{1 - \hat{F}_V(v)}{\hat{f}_V(v)}.$$



Where $\hat{\psi}$ crosses the dashed $\psi = 0$ line is your $\hat{r}$.
For the uniform benchmark $\hat{\psi}(v) = 2v - 1$, so $\hat{r} = \frac{1}{2}$, matching Lecture 5.

Where ψ Comes From: The Seller's Objective

The seller picks a reserve r to maximize expected revenue (seller's own value $= 0$). With one buyer, $V \sim F$, facing a posted price r:

$$\pi(r) = r \cdot \Pr(V \geq r) = r\,[1 - F(r)].$$

Raise r by dr:

- Gain: buyers who still buy each pay dr more: $[1 - F(r)]\,dr$.
- Loss: the marginal buyers (mass $f(r)\,dr$) drop out, each costing r: $rf(r)\,dr$.

$$\pi'(r) = [1 - F(r)] - rf(r) = -f(r)\left[r - \frac{1 - F(r)}{f(r)}\right] = -f(r)\,\psi(r).$$

Setting $\pi'(r) = 0$ gives $\psi(r) = 0$.

- r should be strictly inside the support: there $f(r) > 0$, so $f(r)$ can be divided out and $\pi'(r)$ depends only on $\psi(r)$ (its sign is the opposite of $\psi(r)$'s).
- *Many bidders* (Myerson): expected revenue is $R(r) = \int_r^{\bar{v}} \psi(v)\,dF^n(v)$, so $R'(r) = -\psi(r)\,nF(r)^{n-1}f(r)$.

What If ψ Never Crosses Zero?

A reserve exists only where ψ changes sign on the support. Sometimes it never does:

- $\psi > 0$ everywhere. Even the lowest type has positive virtual value: no interior crossing. The optimal reserve sits at (or below) the bottom of the support — no binding reserve, sell to everyone.
- $\psi < 0$ everywhere. Even the top type's virtual value is negative: the seller never wants to sell, so the "reserve" is above the whole support. (Rare.)

Check the low end: since $F(\underline{v}) = 0$,

$$\psi(\underline{v}) = \underline{v} - \frac{1}{f(\underline{v})},$$

and a reserve binds only if this is negative.

Example

Uniform$[0,1]$: $\psi(0) = 0 - 1 = -1 < 0$, so it binds and crosses at $\frac{1}{2}$.

Uniform$[5,6]$: $\psi(5) = 5 - 1 = 4 > 0$, so no binding reserve.

In code, a bracketing root finder then returns a "no sign change" error — that is the solver telling you there is *no interior reserve*, not a bug.

Pitfalls and a Fallback

- Noisy edges. $\hat{f}_V$ in the denominator is unreliable near the ends of the support, so $\hat{\psi}$ can wiggle there. Keep the bracket inside the *trimmed* region.

- Check before you solve. Confirm $\hat{\psi}(lo) < 0 < \hat{\psi}(hi)$; otherwise the root finder has nothing to bracket.

- No-library fallback. Evaluate $\hat{\psi}$ on a fine grid, find where it flips sign, and linearly interpolate the crossing. This uses only code you already have and should agree with the solver.